Use the Bisection method to find solutions accurate to within $10^{-5}$ for the following problems.

**a.** $x - 2^{-x} = 0 \quad \text{for } 0 \le x \le 1$

In[•]:=

```
a = {0};
b = {1};
p = {a[[1]], a[[1]] + (b[[1]] - a[[1]])/2};
k = 1;
F[x_] := x - 2^(-x);
While[(Abs[p[[k + 1]] - p[[k]]] > 10^(-5)),
 k += 1;
 If[F[p[[k]]] ≠ 0,
  If[Sign[F[p[[k]]]] == -1,
   a = Append[a, p[[k]]];
   b = Append[b, b[[-1]]];, b = Append[b, p[[k]]];
   a = Append[a, a[[-1]]]
  ], Break;
 ];
 p = Append[p, a[[-1]] + (b[[-1]] - a[[-1]])/2];]

m = Table[
   N[{n, a[[n]], b[[n]], p[[n]], F[p[[n + 1]]], Abs[(p[[n + 1]] - p[[n]])/p[[n + 1]]]}, 5], {n, 1, Length[a]}];
Grid[Prepend[m, {"n", "a", "b", "p", "f(p)", "∈a"}], Frame → All]
```

Out[•]=

| n | a | b | p | f(p) | $\in_a$ |
|---|---|---|---|---|---|
| 1.0000 | 0 | 1.0000 | 0 | −0.20711 | 1.0000 |
| 2.0000 | 0.50000 | 1.0000 | 0.50000 | 0.15540 | 0.33333 |
| 3.0000 | 0.50000 | 0.75000 | 0.75000 | −0.023420 | 0.20000 |
| 4.0000 | 0.62500 | 0.75000 | 0.62500 | 0.066571 | 0.090909 |
| 5.0000 | 0.62500 | 0.68750 | 0.68750 | 0.021725 | 0.047619 |
| 6.0000 | 0.62500 | 0.65625 | 0.65625 | −0.00081001 | 0.024390 |
| 7.0000 | 0.64063 | 0.65625 | 0.64063 | 0.010467 | 0.012048 |
| 8.0000 | 0.64063 | 0.64844 | 0.64844 | 0.0048306 | 0.0060606 |
| 9.0000 | 0.64063 | 0.64453 | 0.64453 | 0.0020109 | 0.0030395 |
| 10.000 | 0.64063 | 0.64258 | 0.64258 | 0.00060060 | 0.0015221 |
| 11.000 | 0.64063 | 0.64160 | 0.64160 | −0.00010467 | 0.00076161 |
| 12.000 | 0.64111 | 0.64160 | 0.64111 | 0.00024797 | 0.00038066 |
| 13.000 | 0.64111 | 0.64136 | 0.64136 | 0.000071654 | 0.00019037 |
| 14.000 | 0.64111 | 0.64124 | 0.64124 | −0.000016507 | 0.000095193 |
| 15.000 | 0.64117 | 0.64124 | 0.64117 | 0.000027573 | 0.000047594 |
| 16.000 | 0.64117 | 0.64120 | 0.64120 | $5.5332 \times 10^{-6}$ | 0.000023798 |
| 17.000 | 0.64117 | 0.64119 | 0.64119 | $-5.4870 \times 10^{-6}$ | 0.000011899 |

Find an approximation to $\sqrt{3}$ correct to within $10^{-4}$ using the Bisection Algorithm.

In[•]:=

```
a = {1};
b = {2};
p = {a[[1]], a[[1]] + (b[[1]] - a[[1]])/2};
k = 1;
F[x_] := x^2 - 3;
While[(Abs[p[[k + 1]] - p[[k]]] > 10^(-4)),
 k += 1;
 If[F[p[[k]]] ≠ 0,
  If[Sign[F[p[[k]]]] == -1,
   a = Append[a, p[[k]]];
   b = Append[b, b[[-1]]];, b = Append[b, p[[k]]];
   a = Append[a, a[[-1]]]
  ], Break;
 ];
 p = Append[p, a[[-1]] + (b[[-1]] - a[[-1]])/2];]

m = Table[
   N[{n, a[[n]], b[[n]], p[[n]], F[p[[n + 1]]], Abs[(p[[n + 1]] - p[[n]])/p[[n + 1]]]}, 7], {n, 1, Length[a]}];
Grid[Prepend[m, {"n", "a", "b", "p", "f(p)", "∈a"}], Frame → All]
```

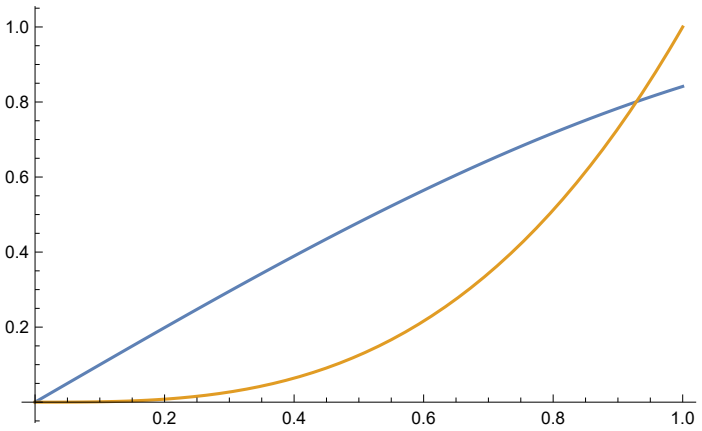
Out[•]=

| n | a | b | p | f(p) | $\in_a$ |
|---|---|---|---|---|---|
| 1.000000 | 1.000000 | 2.000000 | 1.000000 | −0.7500000 | 0.3333333 |
| 2.000000 | 1.500000 | 2.000000 | 1.500000 | 0.06250000 | 0.1428571 |
| 3.000000 | 1.500000 | 1.750000 | 1.750000 | −0.3593750 | 0.07692308 |
| 4.000000 | 1.625000 | 1.750000 | 1.625000 | −0.1523438 | 0.03703704 |
| 5.000000 | 1.687500 | 1.750000 | 1.687500 | −0.04589844 | 0.01818182 |
| 6.000000 | 1.718750 | 1.750000 | 1.718750 | 0.008056641 | 0.009009009 |
| 7.000000 | 1.718750 | 1.734375 | 1.734375 | −0.01898193 | 0.004524887 |
| 8.000000 | 1.726563 | 1.734375 | 1.726563 | −0.005477905 | 0.002257336 |
| 9.000000 | 1.730469 | 1.734375 | 1.730469 | 0.001285553 | 0.001127396 |
| 10.00000 | 1.730469 | 1.732422 | 1.732422 | −0.002097130 | 0.0005640158 |
| 11.00000 | 1.731445 | 1.732422 | 1.731445 | −0.0004060268 | 0.0002819284 |
| 12.00000 | 1.731934 | 1.732422 | 1.731934 | 0.0004397035 | 0.0001409443 |
| 13.00000 | 1.731934 | 1.732178 | 1.732178 | 0.00001682341 | 0.00007047713 |
| 14.00000 | 1.731934 | 1.732056 | 1.732056 | −0.0001946054 | 0.00003523981 |

**5.5** Locate the first nontrivial root of $\sin x = x^3$ where $x$ is in radians. Use a graphical technique and bisection with the initial interval from 0.5 to 1. Perform the computation until $\varepsilon_a$ is less than $\varepsilon_s = 2\%$. Also perform an error check by substituting your final answer into the original equation.

In[•]:=
```
Plot[{Sin[x], x^3}, {x, 0, 1}]
```

Out[•]=



In[•]:=
```
a = {0.8};
b = {1};
p = {a[[1]], a[[1]] + (b[[1]] - a[[1]])/2};
k = 1;
F[x_] := Sin[x] - x^3;
While[(Abs[(p[[k + 1]] - p[[k]]) / p[[k + 1]]] > 0.02),
 k += 1;
 If[F[p[[k]]] ≠ 0,
  If[Sign[F[p[[k]]]] == -1,
   a = Append[a, p[[k]]];
   b = Append[b, b[[-1]]];, b = Append[b, p[[k]]];
   a = Append[a, a[[-1]]]
  ], Break;
 ];
 p = Append[p, a[[-1]] + (b[[-1]] - a[[-1]])/2];]

m = Table[
   N[{n, a[[n]], b[[n]], p[[n]], F[p[[n + 1]]], Abs[(p[[n + 1]] - p[[n]])/p[[n + 1]]]}, 7], {n, 1, Length[a]}];
Grid[Prepend[m, {"n", "a", "b", "p", "f(p)", "∈a"}], Frame → All]
```

Out[•]=

| n | a | b | p | f(p) | $\in_a$ |
|---|---|---|---|---|---|
| 1.000000 | 0.8 | 1.000000 | 0.8 | 0.0543269 | 0.111111 |
| 2.000000 | 0.8 | 0.9 | 0.9 | 0.137155 | 0.0588235 |
| 3.000000 | 0.8 | 0.85 | 0.85 | 0.173032 | 0.030303 |
| 4.000000 | 0.8 | 0.825 | 0.825 | 0.189632 | 0.0153846 |

The fourth-degree polynomial

$$f(x) = 230x^4 + 18x^3 + 9x^2 - 221x - 9$$

has two real zeros, one in $[-1, 0]$ and the other in $[0, 1]$. Attempt to approximate these zeros to within $10^{-6}$ using the

**a.** Method of False Position

In[•]:=
```
a1 = 0;
a2 = 1;
a = {a1};
b = {a2};
p = {0, 1};
k = 2;
F[x_] := 230 x^4 + 18 x^3 + 9 x^2 - 221 x - 9;
While[(Abs[(p[[k]] - p[[k - 1]]) / p[[k]]] > 10^(-6)),
 p = Append[p, a2 - F[a2] (a2 - a1) / (F[a2] - F[a1])];
 k += 1;
 If[F[p[[k]]] ≠ 0,
  If[Sign[F[p[[k]]]] == Sign[F[a2]],
   a2 = p[[k]];
   ,
   a1 = p[[k]];
  ], Break;
 ];
 a = Append[a, a1];
 b = Append[b, a2];
 ]

m = Table[
   N[{n, a[[n]], b[[n]], p[[n]], F[p[[n + 1]]], Abs[(p[[n + 1]] - p[[n]]) / p[[n + 1]]]}, 5], {n, 1, Length[a]}];
Grid[Prepend[m, {"n", "a", "b", "p", "f(p)", "ϵa"}], Frame → All]
```

Out[•]=

| n | a | b | p | f(p) | $\epsilon_a$ |
|---|---|---|---|---|---|
| 1.0000 | 0 | 1.0000 | 0 | 27.000 | 1.0000 |
| 2.0000 | 0.25000 | 1.0000 | 1.0000 | −62.508 | 3.0000 |
| 3.0000 | 0.77376 | 1.0000 | 0.25000 | −83.831 | 0.67690 |
| 4.0000 | 0.94489 | 1.0000 | 0.77376 | −11.265 | 0.18110 |
| 5.0000 | 0.96111 | 1.0000 | 0.94489 | −0.85587 | 0.016882 |
| 6.0000 | 0.96231 | 1.0000 | 0.96111 | −0.061802 | 0.0012417 |
| 7.0000 | 0.96239 | 1.0000 | 0.96231 | −0.0044462 | 0.000089448 |
| 8.0000 | 0.96240 | 1.0000 | 0.96239 | −0.00031978 | $6.4340 \times 10^{-6}$ |
| 9.0000 | 0.96240 | 1.0000 | 0.96240 | −0.000022999 | $4.6274 \times 10^{-7}$ |

**Solve** $5\sin x^2 - 8\cos^5 x = 0$ for the root in the interval [0.5,1.5] by Modified Regula Falsi Method.

In[61]:=
```
a1 = 1;
a2 = 2;
a = {a1};
b = {a2};
p = {a1, a2};
k = 2;
F[x_] := 2 x^3 - 2.5 x - 5;
While[(Abs[(p[[k]] - p[[k - 1]]) / p[[k]]] > 10^(-6)),
 p = Append[p, a2 - F[a2] (a2 - a1) / 2 / (F[a2] / 2 - F[a1])];
 k += 1;
 If[F[p[[k]]] ≠ 0,
  If[Sign[F[p[[k]]]] == Sign[F[a2]],
   a2 = p[[k]];
   ,
   a1 = p[[k]];
  ], Break;
 ];
 a = Append[a, a1];
 b = Append[b, a2];
 ]

m = Table[N[{n, a[[n]], b[[n]], p[[n + 2]], F[p[[n + 2]]], Abs[(p[[n + 2]] - p[[n + 1]]) / p[[n + 2]]]}, 5],
   {n, 1, Length[a] - 1}];
Grid[Prepend[m, {"n", "a", "b", "p", "f(p)", "ϵa"}], Frame → All]
```

Out[70]=

| n | a | b | p | f(p) | $\epsilon_a$ |
|---|---|---|---|---|---|
| 1.0000 | 1.0000 | 2.0000 | 1.64706 | −0.181356 | 0.214286 |
| 2.0000 | 1.64706 | 2.0000 | 1.66718 | 0.0998462 | 0.0120681 |
| 3.0000 | 1.64706 | 1.66718 | 1.66284 | 0.0384648 | 0.00261179 |
| 4.0000 | 1.64706 | 1.66284 | 1.66132 | 0.0171739 | 0.000910523 |
| 5.0000 | 1.64706 | 1.66132 | 1.66068 | 0.00811146 | 0.000388305 |
| 6.0000 | 1.64706 | 1.66068 | 1.66038 | 0.00392758 | 0.000179422 |
| 7.0000 | 1.64706 | 1.66038 | 1.66024 | 0.00192407 | 0.0000859533 |
| 8.0000 | 1.64706 | 1.66024 | 1.66017 | 0.000947902 | 0.0000418869 |
| 9.0000 | 1.64706 | 1.66017 | 1.66013 | 0.000468277 | 0.0000205824 |
| 10.000 | 1.64706 | 1.66013 | 1.66012 | 0.00023165 | 0.000010155 |
| 11.000 | 1.64706 | 1.66012 | 1.66011 | 0.00011467 | $5.02035 \times 10^{-6}$ |
| 12.000 | 1.64706 | 1.66011 | 1.6601 | 0.0000567826 | $2.48438 \times 10^{-6}$ |
| 13.000 | 1.64706 | 1.6601 | 1.6601 | 0.0000281222 | $1.23002 \times 10^{-6}$ |
| 14.000 | 1.64706 | 1.6601 | 1.6601 | 0.000013929 | $6.09137 \times 10^{-7}$ |